

1(a)	$(F =) GMm / x^2$, where G is the (universal) gravitational constant	B1
1(b)(i)	$GMm / x^2 = mv^2 / x$ or $v^2 = GM / x$	C1
	$v^2 = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) / (3.4 \times 10^6 + 240 \times 10^3)$ so $v = 3.7 \times 10^3 \text{ m s}^{-1}$	A1
1(b)(ii)	potential energy = $(-)GMm / x$	C1
	$E_A = (-)(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (3.64 \times 10^6)$ or $E_B = (-)(6.67 \times 10^{-11} \times 7.5 \times 10^{23} \times 650) / (5.00 \times 10^7)$	M1
	correct substitution and subtraction $E_B - E_A$ shown, leading to $\Delta E_p = 8.3 \times 10^9 \text{ J}$	A1
	or	
	$\phi = (-)GM / x$ and potential energy = $m\phi$	(C1)
	$\Delta\phi = (6.67 \times 10^{-11} \times 7.5 \times 10^{23}) \times [(1 / (3.64 \times 10^6)) - (1 / (5.00 \times 10^7))]$ $(= 1.27 \times 10^7 \text{ J kg}^{-1})$	(M1)
	$\Delta E_p = 1.27 \times 10^7 \times 650$ $= 8.3 \times 10^9 \text{ J}$	(A1)
1(c)	kinetic energy <u>or</u> potential energy decreases	B1
	kinetic energy <u>and</u> potential energy decrease so total energy decreases	B1